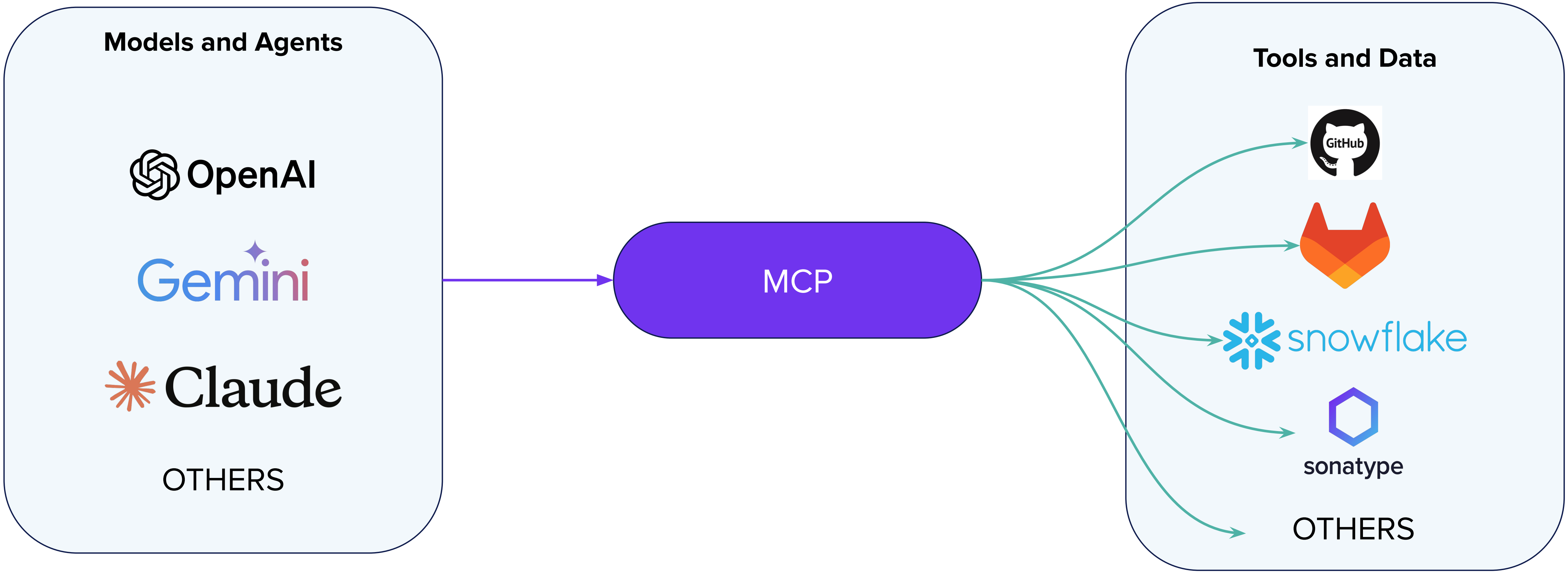
MCP at Scale: Unlocking AI-Driven Innovation

Model Context Protocol Servers



Why MCP?

LLMs excel at language-oriented tasks

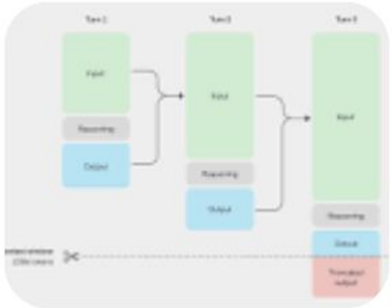
- Translation
- Summarization
- **Code generation**

They struggle with

- Reasoning
- Inference

AI Overview

LLMs can perform reasoning and inference by pattern recognition on vast datasets but require techniques like "chain-of-thought" prompting, inference-time compute scaling, and fine-tuning to enhance performance on complex tasks like logical deduction and multi-step problems. Current LLMs often struggle with deep commonsense and causal reasoning, displaying "jagged intelligence" where advanced skills coexist with significant weaknesses, though ongoing research focuses on models with more inherent planning and reasoning abilities.



An Example: Dependency Selection

"I need a javascript PDF library that can both import and export PDFs and is compatible with PDFs generated by both Acrobat and the Mac Preview app."

PDF-LIB

Create and Modify

Create PDF documents from scratch, or modify existing PDF documents. Draw text, images, and vector graphics. Embed your own fonts. Even embed and draw pages from other PDFs.

Generating PDF files based on template

Help

Hey all,

Fairly new to this field and was wondering how I can generate PDF files using a template that I've created in Adobe Acrobat. I've left placeholder values where I want the dynamic data to be (below is a sc

aXenDeveloper · 1y ago

https://pdf-lib.js.org/

3 Reply ...

An Example: Dependency Selection

"I need a javascript PDF library that can both import and export PDFs and is compatible with PDFs generated by both Acrobat and the Mac Preview app ***that is actively maintained****."*

ChatGPT 5

pdf-lib (Hopding/pdf-lib)	Moderately active. Maintains a presence, with documentation current. However, I saw a concern: there is an open issue "Is this thing still on?" from ~ March 2023 where someone noted no recent commits. GitHub Also, the npm side shows version history. GitHub +1

😞

pdf-lib TS

1.17.1 • Public • Published 4 years ago

End Of Life

Factors to consider

- Dev activity
- Issue tracker activity
- Discussion forum chatter
- Release patterns
- Number of active developers
- Download patterns

All this should be synthesized into a determination and ideally reviewed by human reviewers.

Things to incorporate

Project descriptions / documentation ✔

Meta-data analysis ✘

Curated data ✘

- What if we add: “Are **not malicious**?”
  - Malicious releases are designed to sound legitimate and look appealing. Relying solely on project descriptions is misleading.
  - Determined via: Metadata analysis + human review

Things to incorporate

Project descriptions / documentation ✔

Meta-data analysis ✘

Curated data ✘

Reasoning and optimization ✘

- What if we add: “**Prefer more popular** libraries”
  - This layers on **reasoning and optimization**
  - Which factors are hard vs soft constraints?
  - How should popularity and other soft constraints be weighted?

What Are We Trying to Solve

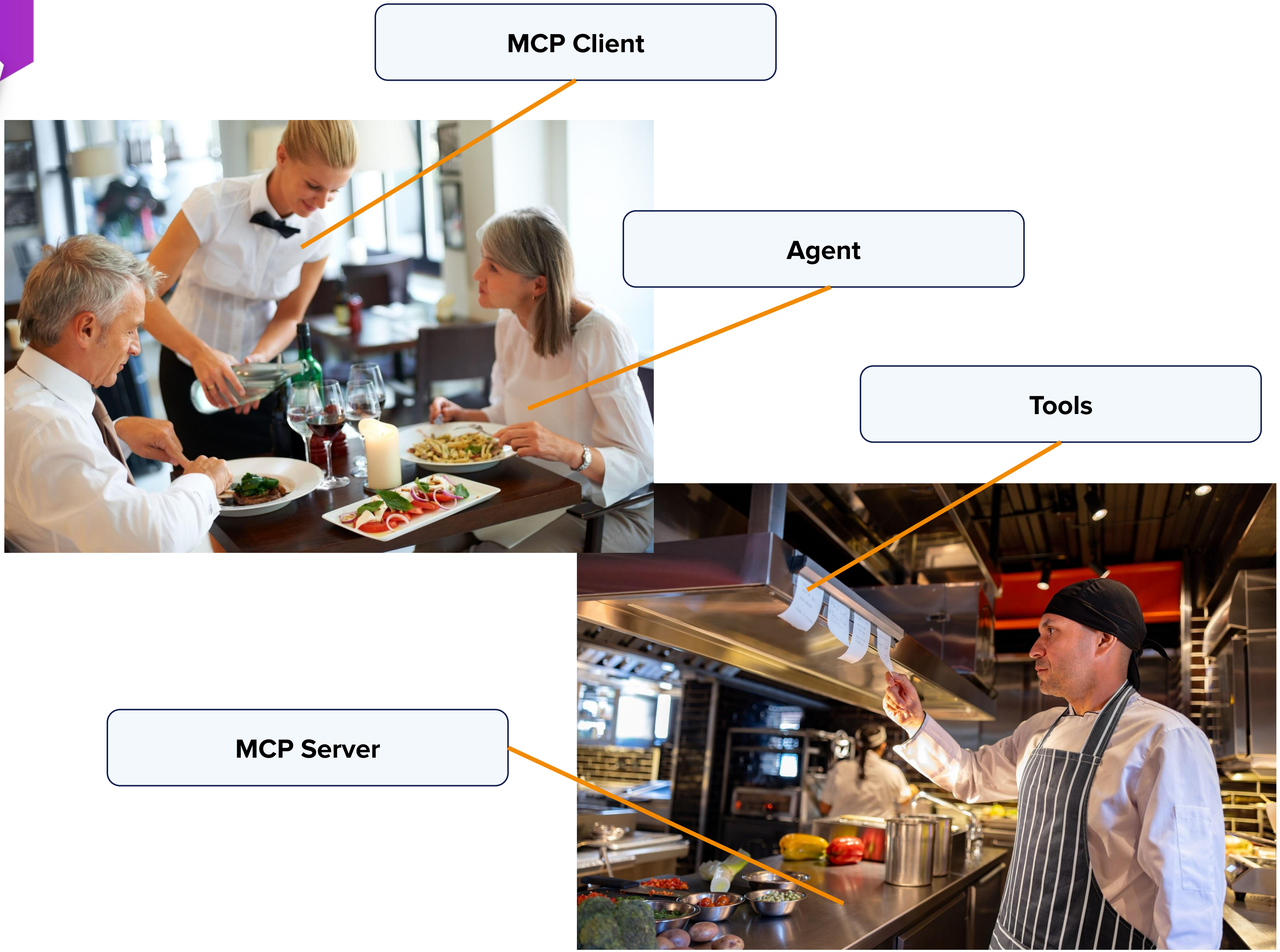
Knowledge Limitations
LLMs rely on training data that can quickly become outdated or lack full context. They lack **up-to-date curated data**.

Domain Knowledge Gaps
LLMs lack deep understanding of specialized domains especially in areas of Private or Unique data assets. They struggle when synthesizing data to to **infer domain-specific attributes**.

Reasoning Gaps
LLMs struggle to **explore** complex search spaces **and optimize** across multiple dimensions.

Integrations
Current methods for connecting LLMs to **external data sources** can be frustrating and costly without a standard.

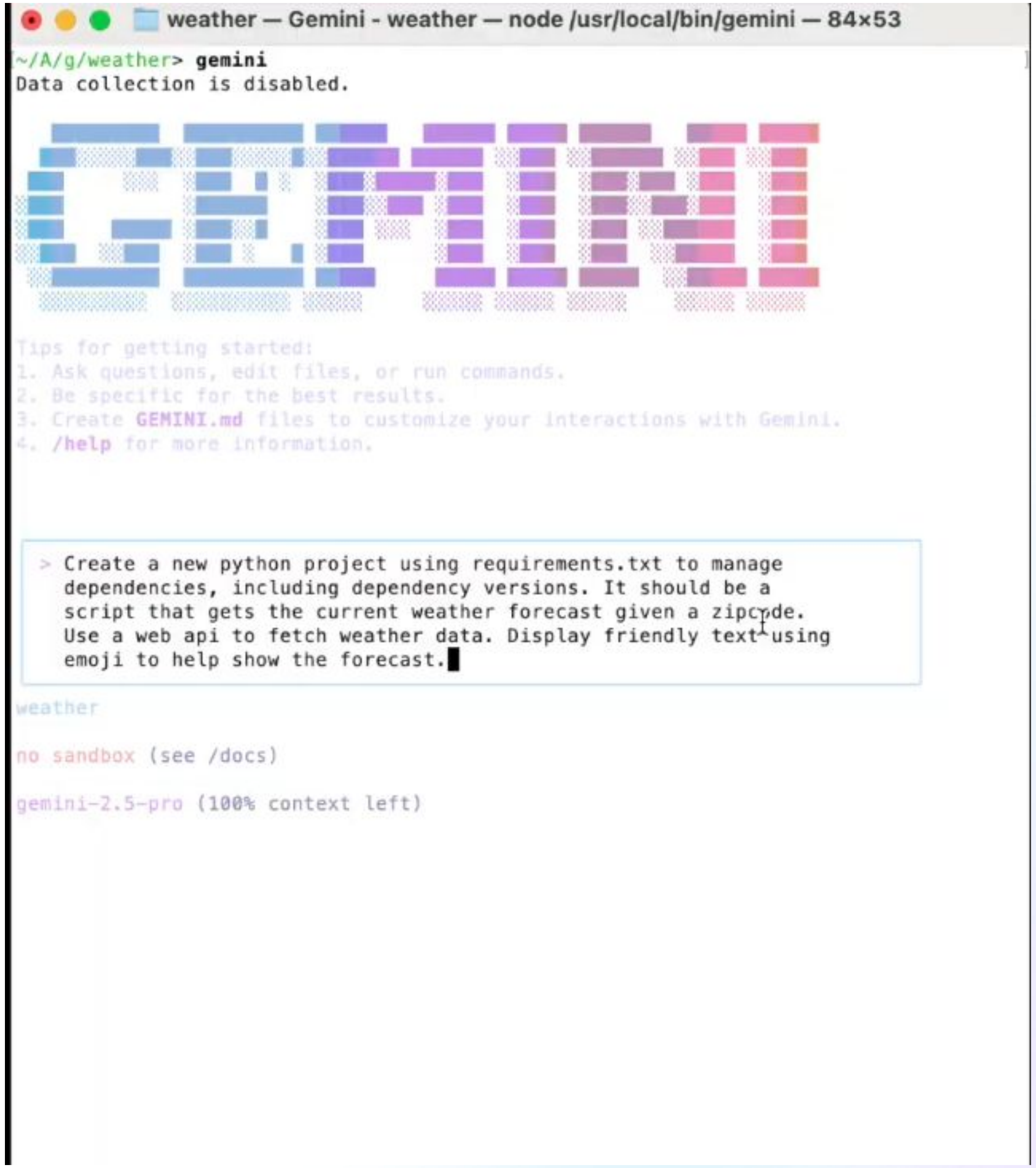
MCP as a Restaurant



MCP as a Domain “Expert”

Let’s Build a Weather App

Prompt: Create a new python project using requirements .txt to manage dependencies including dependency version. It should be a script that gets the current weather forecast given a zipcode. Use a web api to fetch weather data. Display friendly text using emoji to help show the forecast.



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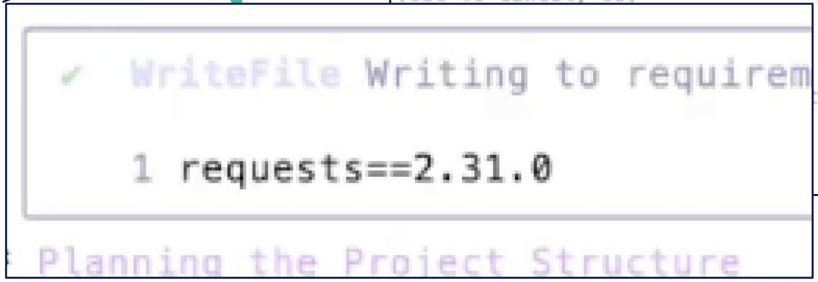

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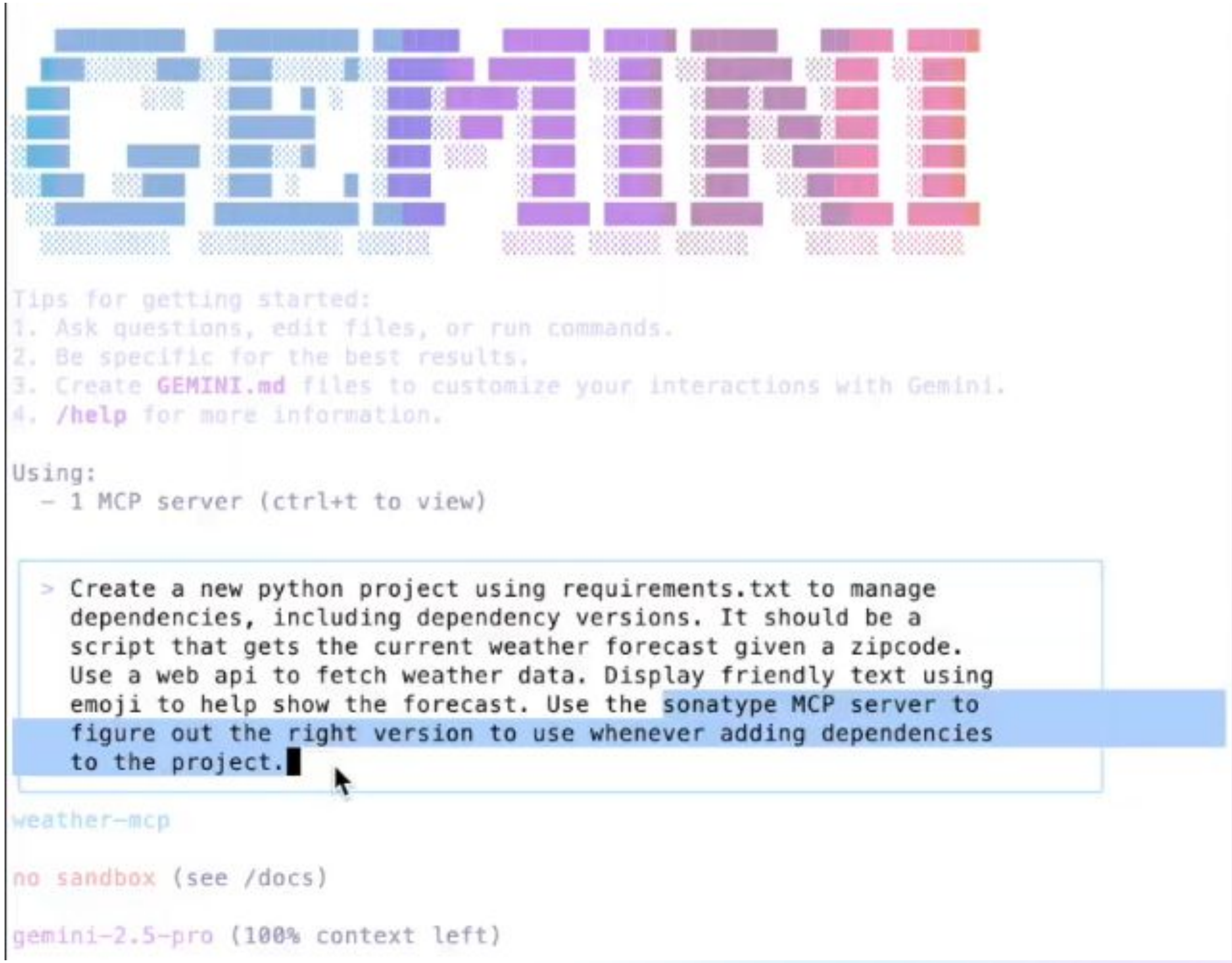

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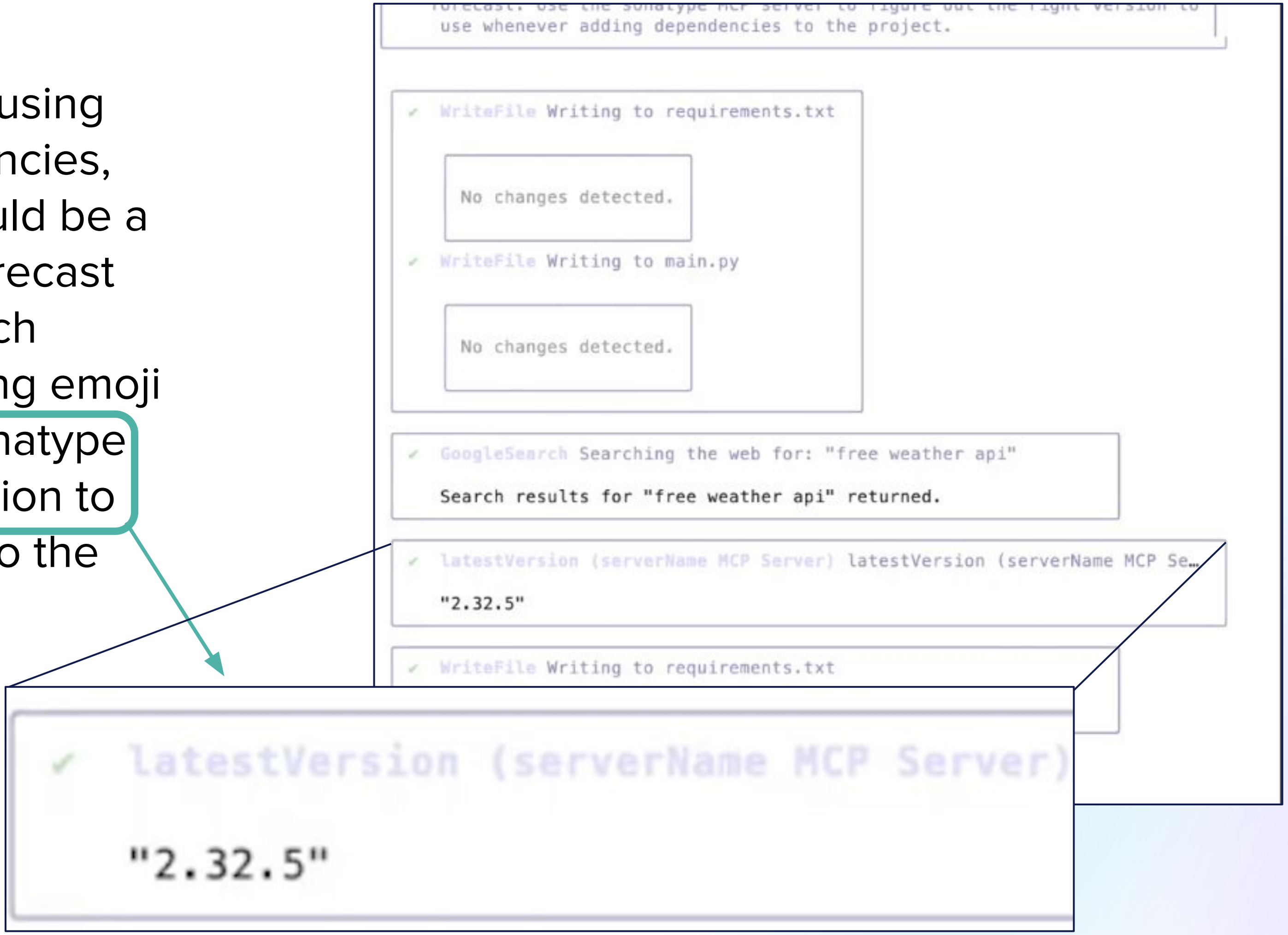

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MCP Types

- Interfaces / actions
  - GitHub: Create a pull request
  - Jira: Create an issue
  - Jenkins: Trigger a build
- Data
  - Elastic Search
  - MongoDB
  - Sonatype MCP
- Reasoning
  - Next frontier...

OWASP LLM Top 10

What do they mean?

1. Prompt Injection
2. Sensitive Info Disclosure.
3. Supply Chain
4. Data & Model Poisoning
5. Improper Output Handling.
6. Excessive Agency
7. System Prompt Leakage.
8. Vector & Embed Weaknesses
9. Misinformation
10. Unbounded Consumption

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OWASP LLM Top 10 - Definitions Part I

Prompt Injection
Certain inputs can cause models to produce harmful outputs, such as bypassing safety protocols or revealing sensitive info.

Sensitive Info Disclosure
Models that are improperly configured can inadvertently disclose sensitive or private info, like accessing confidential databases.

Supply Chain
AI models and components carry the same risks as open-source components in terms of vulnerabilities, malware, licenses.

Data & Model Poisoning
Bad actors can alter the data or the AI models themselves, which can lead to harmful attacks or biased outputs.

Improper Output Handling
AI models can generate unintended outputs (e.g., NSFW content) and must be scrutinized before being integrated into systems.

OWASP LLM Top 10 - Definitions Part II

Excessive Agency	AI can have too much autonomy to execute jobs, leading to harmful actions or poor decisions without proper oversight.
System Prompt Leakage	The unintended exposure of internal model workings, potentially giving attackers insights into how to exploit the model.
Vector & Embed Weaknesses	Weak vectors/embeddings can be exploited to inject harmful content, manipulate outputs, or access sensitive information.
Misinformation	AI models can generate convincing but false information, risking security breaches, lawsuits, and reputational harm.
Unbounded Consumption	User/attackers can overload AI systems with high input volumes, causing poor performance and unexpected costs/financial strain.

Want to continue the conversation?

Come by our Booth to schedule a session for a follow-up with a Product Expert

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